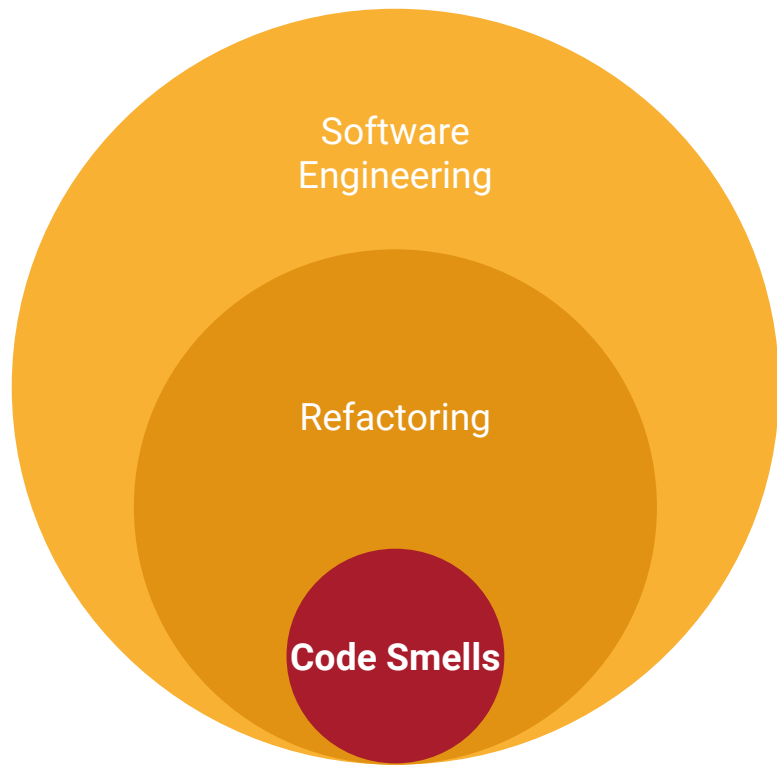


Inspiration: Code Smells



"A code smell is
a surface indication
that usually corresponds to a
deeper problem in the system."
([Fowler 2006](#))

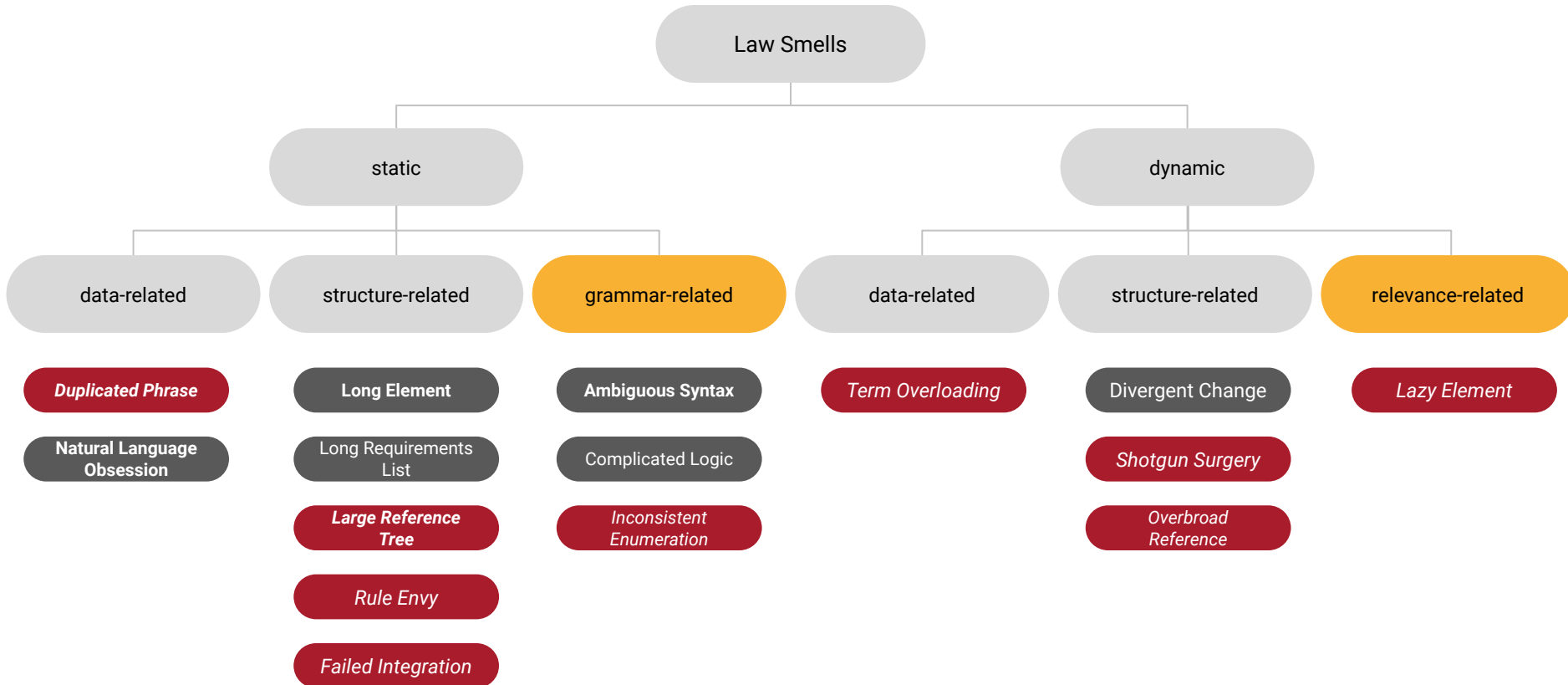
Adaptation: Law Smells

Law Smell Profile		
#	Category	Question
1	Description	What is it?
2	Problem	Why is it problematic?
3	Detection	How do we detect it?
4	Mitigation	If we decide that it needs mitigation, how can we mitigate it?
5	Example	For illustration, how does it manifest in practice?

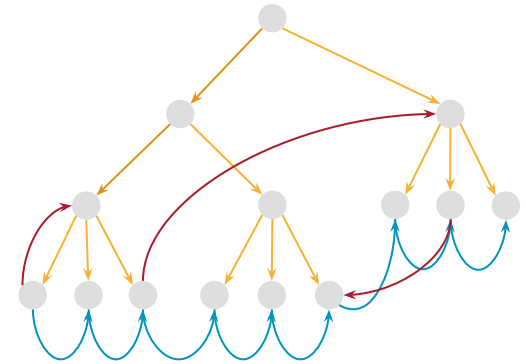
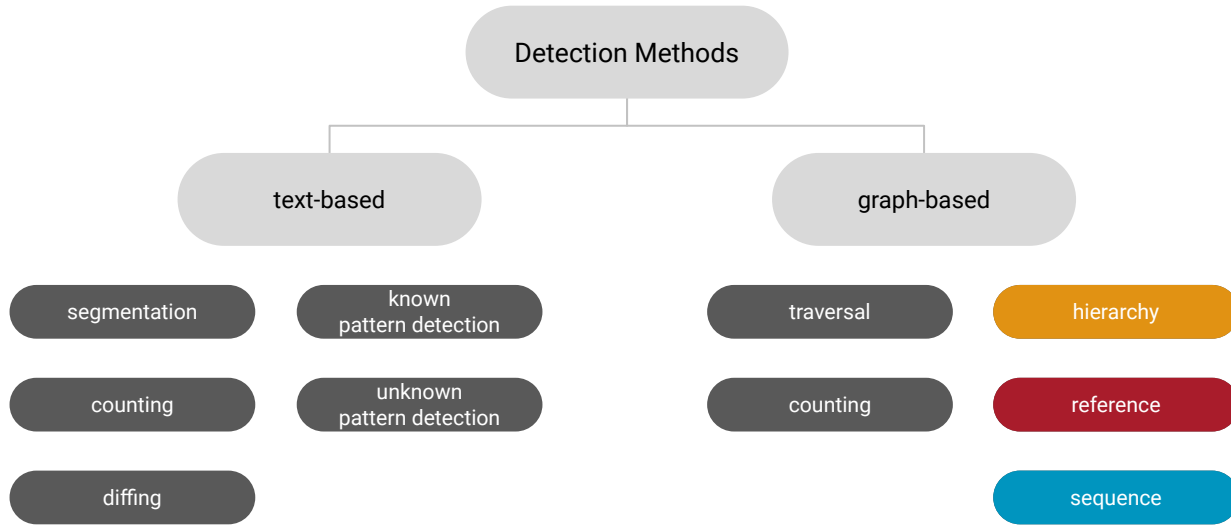
Implementation I: Distinctions

	Categories	Question
Stability	static dynamic	Can it be detected by looking at the text alone?
Modality	data structure grammar relevance	What aspect of the law does it relate to?
Locality	local non-local	Can it be detected by looking at a single element of law?

Implementation II: Taxonomy



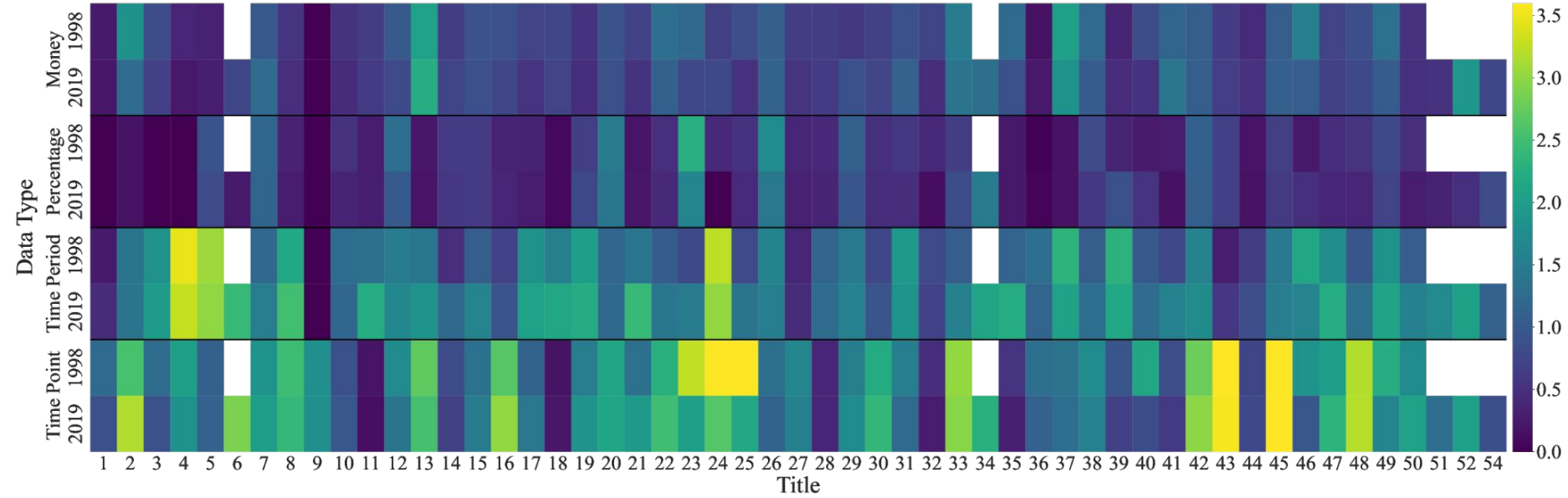
Implementation III: Detection



USC Demo: Duplicated Phrase

Duplicated Phrase in Title 6 – Domestic Security	# Occurrences
[[director] of the] bureau of citizenship and immigration services	[18,66]
natural disasters, acts of terrorism, (and or) other man-made disasters	[11,52]
the committee on homeland security (and governmental affairs of the senate of the house of representatives)	[30,31]
cyber threat indicators (and or) defensive measures	[20,26]
information within the scope of the information sharing environment, including homeland security information, terrorism information, and weapons of mass destruction information	[26,26]

USC Demo: Natural Language Obsession



USC Demo: Long Element

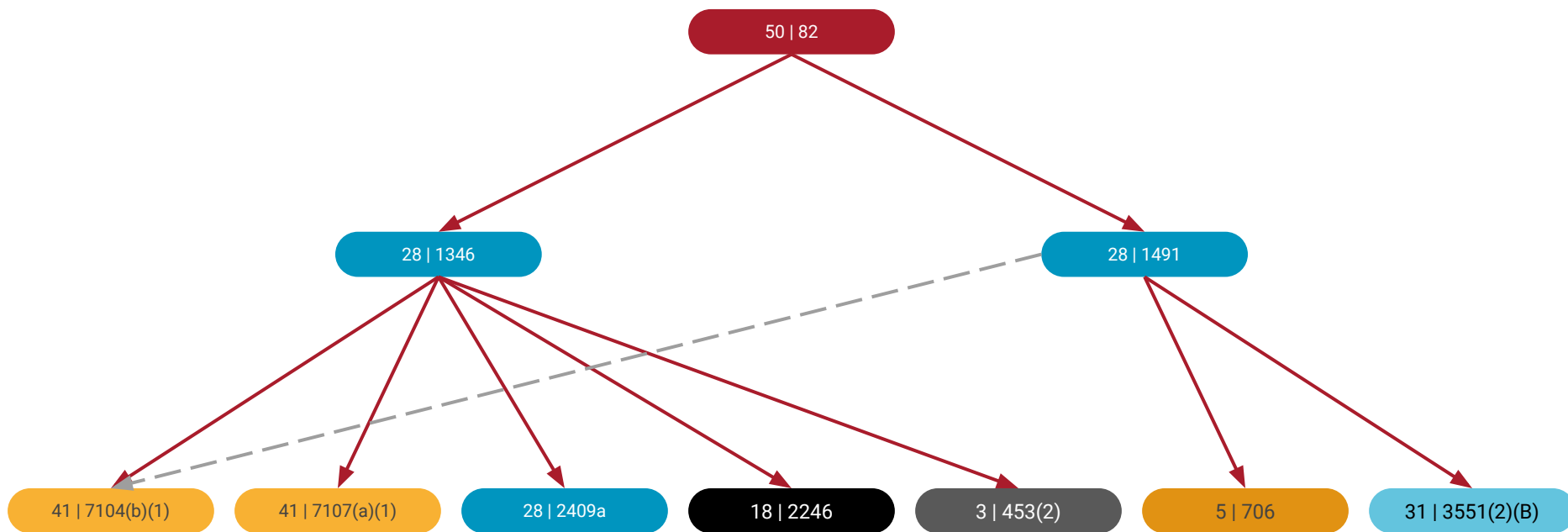


United States Code

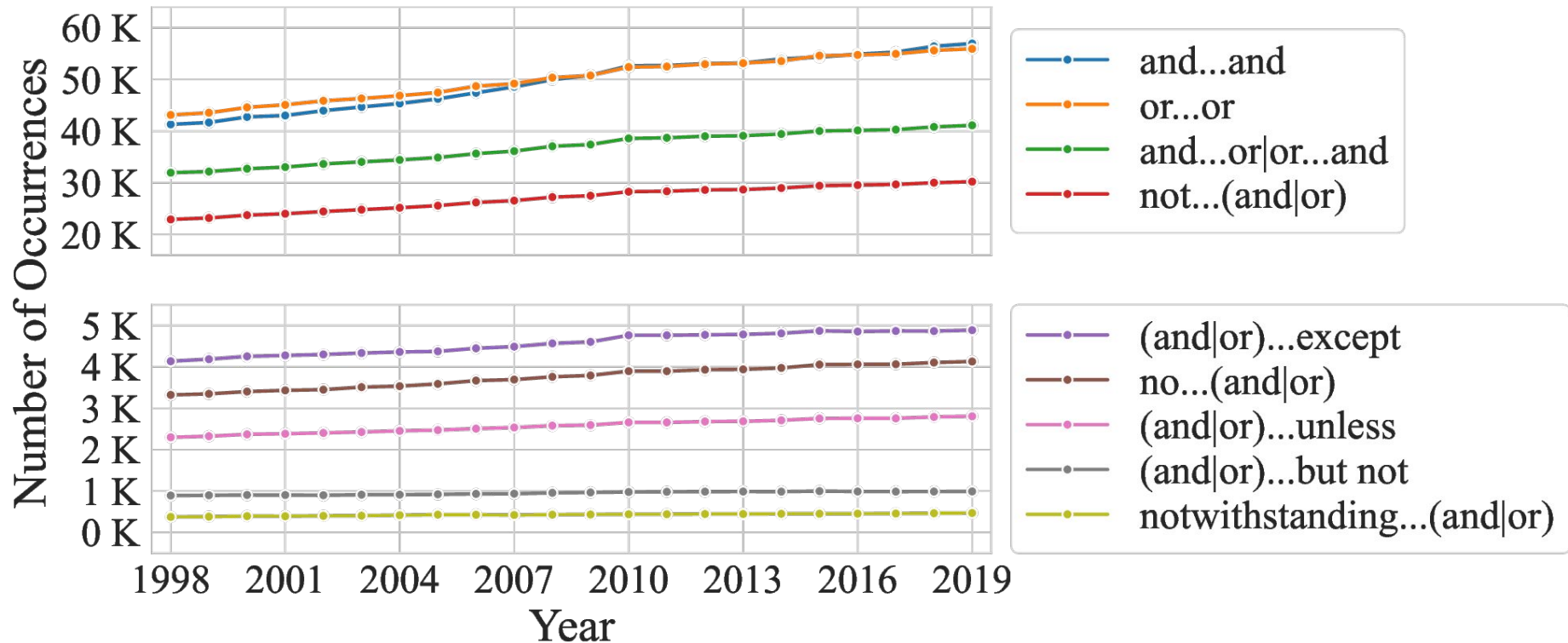
Title 15: Commerce and Trade

Chapter 14A: Aid to Small Business

USC Demo: Large (Diversity) Reference Tree



USC Demo: Ambiguous Syntax



Outlook: Legal Refactoring

Maintainability

Integrated Development
Environments

Version Control

Separating Frontend
and Backend

Code Climate

Usability

Interactive User
Interfaces

Impact Analysis

Check out our full paper!

<https://arxiv.org/abs/2110.11984>